

CUSTOMER SEGMENTATION PROJECT

K-Means Clustering | Python (scikit-learn) | FY 2024

Internship Task 2 | 500 Customers | 4 Segments Identified

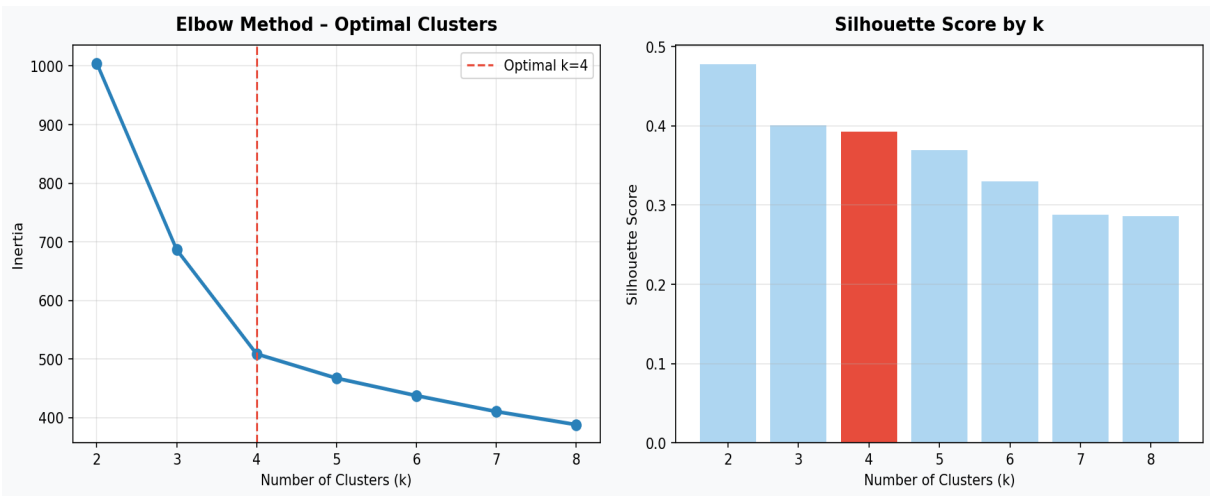
1. Project Overview

This project applies **K-Means clustering** to segment 500 customers based on behavioral and demographic features: Age, Annual Income, Monthly Spend, Purchase Frequency, and Tenure. The analysis identifies 4 distinct segments to enable targeted marketing strategies and personalized customer experiences.

2. Methodology

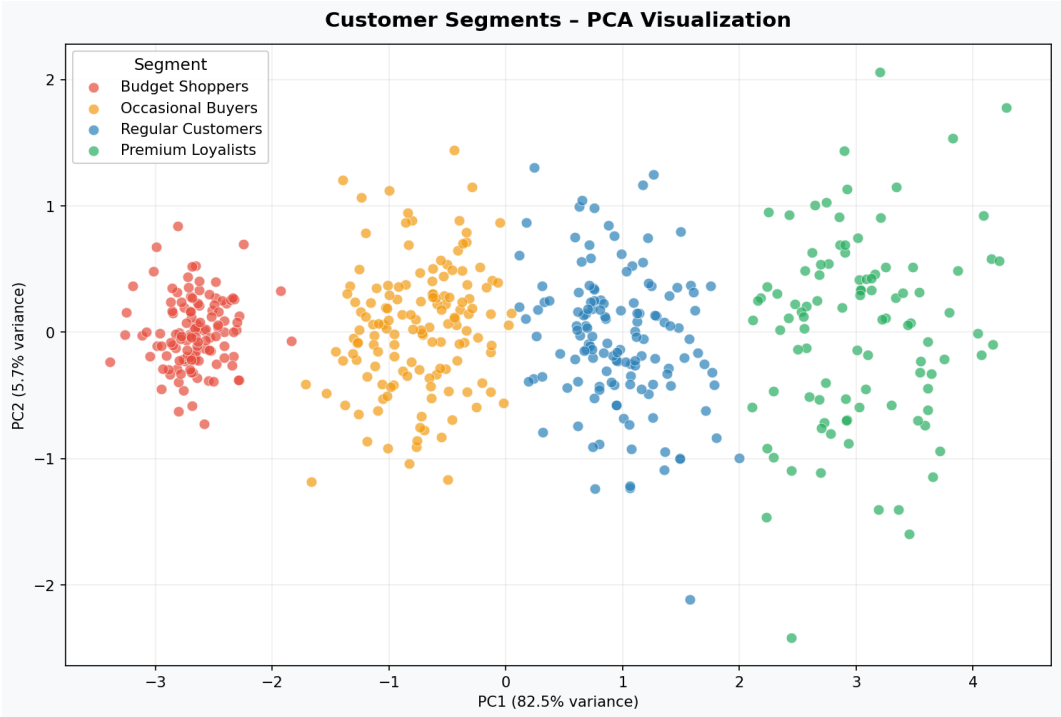
Step	Technique	Tool
Data Generation	Synthetic dataset – 500 customers, 9 features	Python / NumPy
Preprocessing	StandardScaler normalization	scikit-learn
Optimal k	Elbow Method + Silhouette Score	scikit-learn
Clustering	K-Means (k=4, n_init=10)	scikit-learn
Dimensionality Reduction	PCA (2 components for visualization)	scikit-learn
Visualization	Scatter, Bar, Pie, Profile charts	Matplotlib / Seaborn

3. Elbow Method & Silhouette Analysis



The Elbow Method shows a clear inflection at **k=4**, and the Silhouette Score confirms this with a score of **0.393**, indicating well-separated and cohesive clusters.

4. PCA Cluster Visualization

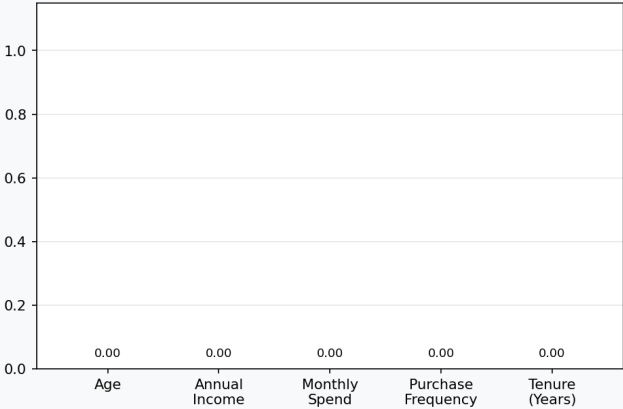


PCA reduces the 5-dimensional feature space to 2 principal components, capturing the major variance. The plot clearly shows **4 well-separated clusters**, validating the K-Means result.

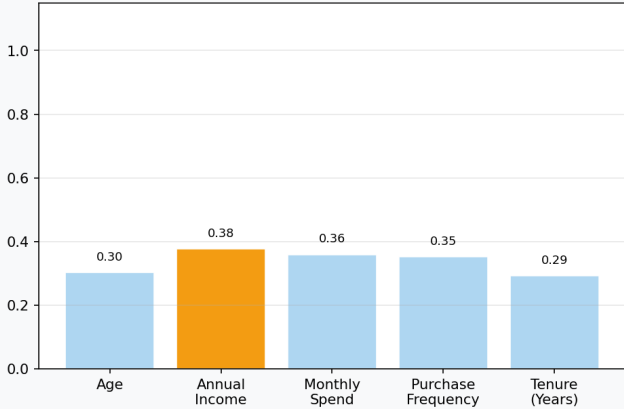
5. Segment Profiles

Segment Profiles - Feature Comparison

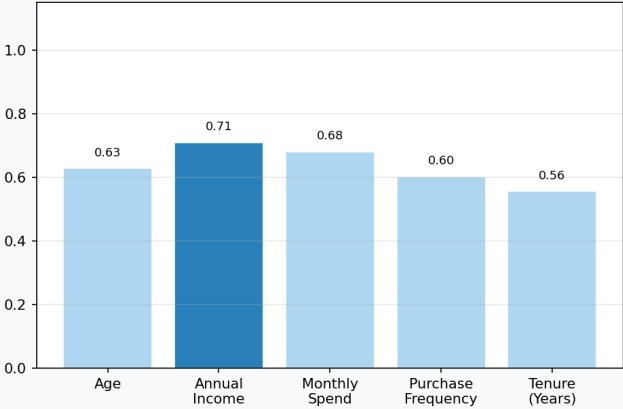
Budget Shoppers



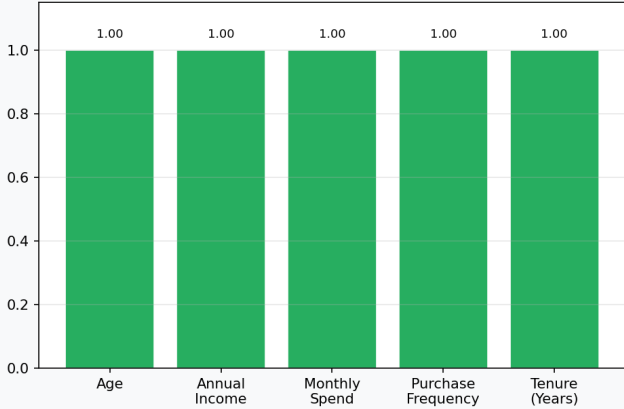
Occasional Buyers



Regular Customers



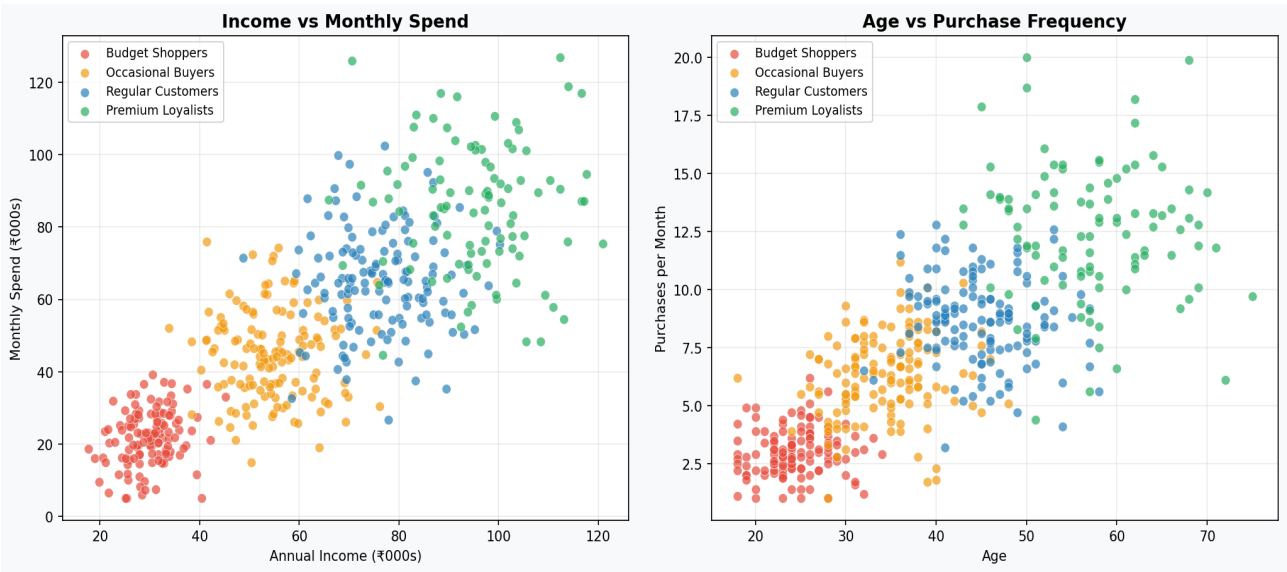
Premium Loyalists



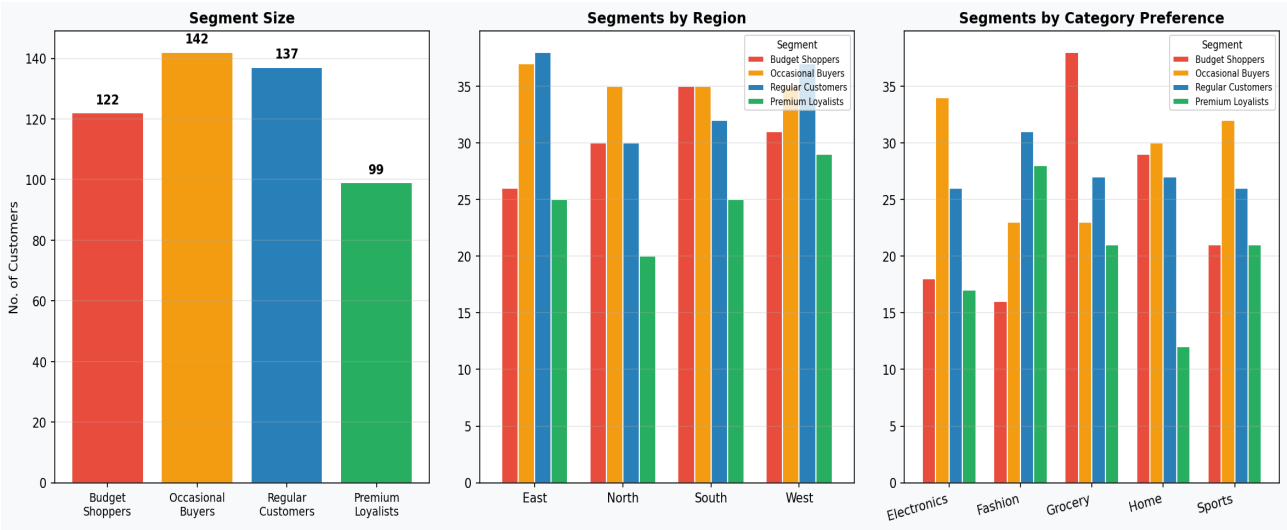
6. Key Segment Statistics

Segment	Count	Avg Age	Avg Income	Avg Spend/Mo	Avg Freq	Avg Tenure
■ Budget Shoppers	122	24.4	■29,922	■21,800	3.0/mo	0.9 yrs
■ Occasional Buyers	142	34.3	■54,584	■44,600	6.3/mo	3.0 yrs
■ Regular Customers	137	45.0	■76,372	■65,000	8.7/mo	4.9 yrs
■ Premium Loyalists	99	57.2	■95,353	■85,300	12.4/mo	8.1 yrs

7. Income vs Spend & Age vs Frequency



8. Segment Distribution



9. Business Recommendations

■ Budget Shoppers (24%)	Young (avg 24), low income. Target with discounts, student offers, EMI schemes, and entry-level product bundles. Focus on Electronics and Fashion categories.
■ Occasional Buyers (28%)	Mid-income, moderate engagement. Re-engage with loyalty points, push notifications, weekend flash sales. High conversion potential if nurtured correctly.
■ Regular Customers (27%)	Stable, high-value segment. Reward with membership perks, early access to sales, and personalized recommendations based on purchase history.
■ Premium Loyalists (20%)	Oldest, highest income and spend. Offer VIP programs, concierge service, premium product lines, and exclusive events. Protect churn at all costs — highest LTV segment.